23/01/2025
Morning

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Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1
[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Let $\left|\frac{\bar{z}-i}{2\bar{z}+i}\right|=\frac{1}{3}$, $z \in C$, be the equation of a circle with center at C. If the area of the triangle, whose vertices are at the points (0, 0), C and (α, 0) is 11 square units, then α^2 equals:

(1) $\frac{81}{25}$ (2) 50

(3) 100 (4) $\frac{121}{25}$

Answer (3)

Sol. Let $z = x + iy \Rightarrow \bar{z} = x - iy$

$3|\bar{z}-i| = 1|2\bar{z}+i|$

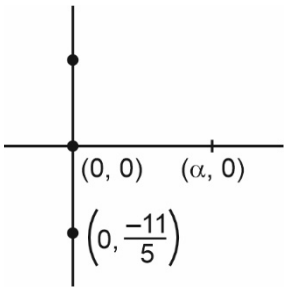
$= 3|(x-(y+1)i| = |2x+i(1-2y)|$

$= 3\sqrt{x^2+(y+1)^2} = \sqrt{(2x)^2+(1-2y)^2}$

$= 9(x^2+y^2+2y+1) = 4x^2+4y^2-4y+1$

$\Rightarrow 5x^2+5y^2+22y+8=0$

$\Rightarrow$ Centre $\equiv \left(0,-\frac{11}{5}\right)$



Area of Δ

$= \frac{1}{2}|\alpha|\left|\frac{-11}{5}\right| = 11$

$\Rightarrow |\alpha| = 10$

$\Rightarrow \alpha^2 = 100$

2. Let $f(x) = \log_e x$ and $g(x)=\frac{x^4-2x^3+3x^2-2x+2}{2x^2-2x+1}$.

Then the domain of fog is

(1) $[1, \infty)$ (2) $(0, \infty)$

(3) $\mathbb{R}$ (4) $[0, \infty)$

Answer (3)

Sol. $f(g(x)) = \ln\left(\frac{x^4-2x^3+3x^2-2x+2}{2x^2-2x+1}\right)$

Since $2x^2-2x+1>0 \quad \forall x \in \mathbb{R} \because (-2)^2-4(2)<0$

Consider

$x^4-2x^3+3x^2-2x+2$

$= (x^4-2x^3+x^2)+(x^2-2x+1)+(1+x^2)$

$= x^2(x-1)^2+(x-1)^2+(x^2+1)>0 \forall x \in \mathbb{R}$

$\Rightarrow g(x)>0 \ \forall x \in \mathbb{R}$

$\Rightarrow \ln f((x)),\ f(x)>0 \ \forall x \in \mathbb{R}$

$\Rightarrow x \in \mathbb{R}$ is domain

3. The number of words, which can be formed using all the letters of the word "DAUGHTER", so that all the vowels never come together, is

(1) 35000 (2) 37000

(3) 34000 (4) 36000

Answer (4)

Sol. E : all vowel never come together

$\bar{E}$: all vowel come together

$n(E) = n(U) - n(\bar{E})$ $\boxed{AUE}$ DGHTR

= 8! – (6!)(3!)

= 6!(7 × 8 – 6) = 720 (56 – 6)

= 720 × 50 = 36000



4. One die has two faces marked 1, two faces marked 2, one face marked 3 and one face marked 4. Another die has one face marked 1, two faces marked 2, two faces marked 3 and one face marked 4. The probability of getting the sum of numbers to be 4 or 5, when both the dice are thrown together, is

(1) $\frac{2}{3}$ (2) $\frac{4}{9}$

(3) $\frac{3}{5}$ (4) $\frac{1}{2}$

Answer (4)

Sol. Die 1 : {1, 1, 2, 2, 3, 4}

Die 2 : {1, 2, 2, 3, 3, 4}

(die 1, die 2) = {(1, 3), (3, 1), (2, 2), (2, 3), (3, 2), (1, 4), (4, 1)}

$$=\frac{2}{6}\times\frac{2}{6}+\frac{1}{6}\times\frac{1}{6}+\frac{2}{6}\times\frac{2}{6}+\frac{2}{6}\times\frac{2}{6}+\frac{1}{6}$$

$$\times\frac{2}{6}+\frac{2}{6}\times\frac{1}{6}+\frac{1}{6}\times\frac{2}{6}=\frac{1}{2}$$

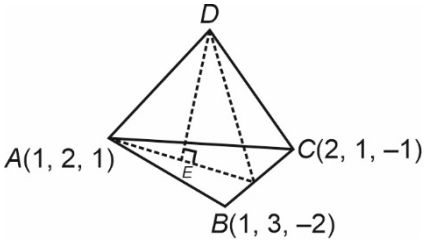
5. Let the position vectors of the vertices A, B and C of a tetrahedron $ABCD$ be $\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+3\hat{j}-2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median line segment through A of the triangle ABC at the point E. If the length of AD is $\frac{\sqrt{110}}{3}$ and the volume of the tetrahedron is $\frac{\sqrt{805}}{6\sqrt{2}}$, then the position vector of E is

(1) $\frac{1}{12}(7\hat{i}+4\hat{j}+3\hat{k})$ (2) $\frac{1}{6}(7\hat{i}+12\hat{j}+\hat{k})$

(3) $\frac{1}{6}(12\hat{i}+12\hat{j}+\hat{k})$ (4) $\frac{1}{2}(\hat{i}+4\hat{j}+7\hat{k})$

Answer (2)

Sol.



Volume = $\frac{1}{3}\times\text{Base}\times h=\frac{\sqrt{805}}{6\sqrt{2}}$

Base area = $\frac{1}{2}\left|\overline{AB}\times\overline{AC}\right|$

$$=\frac{1}{2}\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\0&1&-3\\1&-1&-2\end{vmatrix}=\frac{1}{2}\left|5\hat{i}+3\hat{j}+\hat{k}\right|=\frac{1}{2}\sqrt{35}$$

$$\frac{1}{3}\times\frac{1}{2}\sqrt{35}\times h=\frac{\sqrt{805}}{6\sqrt{2}}$$

$$\Rightarrow\quad h=\sqrt{\frac{23}{2}}$$

Now $AE^2 = AD^2 - DE^2$

$$=\frac{110}{9}-\frac{23}{2}=\frac{13}{18}$$

$$AE=\sqrt{\frac{13}{18}}$$

$$\overline{AE}=|AE|\left(\frac{\hat{i}-5\hat{k}}{\sqrt{26}}\right)=\frac{\hat{i}-5\hat{k}}{6}$$

Position vector of E = $\frac{\hat{i}-5\hat{k}}{6}+\hat{i}+2\hat{j}+\hat{k}$

$$=\frac{1}{6}(7\hat{i}+12\hat{j}+\hat{k})$$

6. Let the area of a ΔPQR with vertices $P(5, 4)$, $Q(-2, 4)$ and $R(a, b)$ be 35 square units. If its orthocentre and centroid are $O\left(2, \frac{14}{5}\right)$ and $C(c, d)$ respectively, then $c + 2d$ is equal to

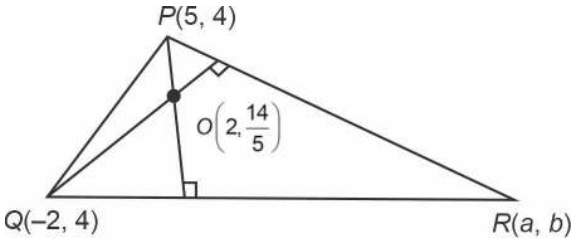
(1) $\frac{7}{3}$ (2) 2

(3) 3 (4) $\frac{8}{3}$

Answer (3)



Sol.



$QR: y-4=-\left(\frac{5-2}{4-\frac{14}{5}}\right)(x+2)$

$QR: 5x+2y+2=0 \quad \ldots(i)$

$PR: y-4=\frac{4}{\frac{14}{5}-4}(x-4)$

$PR: 10x-3y-38=0 \quad \ldots(ii)$

Solving (i) & (ii)

$R\ (2,-6)$

Centroid $C=\left(\frac{5-2+2}{3},\frac{4+4-6}{3}\right)\equiv\left(\frac{5}{3},\frac{2}{3}\right)$

$\Rightarrow \quad c=\frac{5}{3},\ d=\frac{2}{3}$

$c+2d=\frac{5}{3}+\frac{4}{3}=3$

7. Let $R=\{(1,2),(2,3),(3,3)\}$ be a relation defined on the set {1, 2, 3, 4}. Then the minimum number of elements, needed to be added in R so that R becomes an equivalence relation, is:

(1) 8 (2) 7

(3) 9 (4) 10

Answer (2)

Sol. $R=\{(1,2),(2,3),(3,3)\}$ be relation defined on set {1,2,3,4} then the elements added to make this equivalence relation are (1,1), (2,2), (2,1), (3,2), (3,1), (1,3), (4, 4).

Total number of elements added = 7.

8. Let $I(x)=\int\frac{dx}{(x-11)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}}$. If $I(37)-I(24)=\frac{1}{4}\left(\frac{1}{b^{\frac{1}{13}}}-\frac{1}{c^{\frac{1}{13}}}\right)$, $b,\ c\in N$, then $3(b+c)$ is equal to

(1) 22 (2) 40

(3) 26 (4) 39

Answer (4)

Sol. $I(x)=\int\frac{dx}{(x-11)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}}$

$=\int\frac{dx}{(x-11)^2\left(\frac{x+15}{x-11}\right)^{\frac{15}{13}}}$

Let $\frac{x+15}{x-11}=t\Rightarrow\frac{-26}{(x-11)^2}dx=dt$

$\therefore \quad I(x)=\int\frac{-\frac{1}{26}dt}{t^{\frac{15}{13}}}$

$=\frac{1}{4}t^{-\frac{2}{13}}+c=\frac{1}{4}\left(\frac{x-11}{x+15}\right)^{\frac{2}{13}}+c$

$\therefore \quad I(37)-I(24)=\frac{1}{4}\left(\left(\frac{1}{2}\right)^{\frac{2}{13}}-\left(\frac{1}{3}\right)^{\frac{2}{13}}\right)$

$\therefore \quad b=4,\ c=9\Rightarrow 3(b+c)=39$

9. If the system of equations

$(\lambda-1)x+(\lambda-4)y+\lambda z=5$

$\lambda x+(\lambda-1)y+(\lambda-4)z=7$

$(\lambda+1)x+(\lambda+2)y-(\lambda+2)z=9$

has infinitely many solutions, then $\lambda^2+\lambda$ is equal to

(1) 12 (2) 10

(3) 20 (4) 6

Answer (1)



Sol. The given system of equation are

$(\lambda - 1)x + (\lambda - 4)y + \lambda z = 5$

$\lambda x + (\lambda - 1)y + (\lambda - 4)z = 7$

$(\lambda + 1)x + (\lambda + 2)y - (\lambda + 2)z = 9$

has infinite many solutions, then

$D = D_1 = D_2 = D_3 = 0$

$$\therefore \begin{vmatrix} \lambda-1 & \lambda-4 & \lambda \\ \lambda & \lambda-1 & \lambda-4 \\ \lambda+1 & \lambda+2 & -(\lambda+2) \end{vmatrix} = 0$$

$$\text{or, } \begin{vmatrix} -1 & -3 & 4 \\ -1 & -3 & 2\lambda-2 \\ \lambda+1 & \lambda+2 & -\lambda-2 \end{vmatrix} = 0$$

$$\text{or, } \begin{vmatrix} -1 & -3 & 4 \\ 0 & 0 & 2\lambda-6 \\ \lambda+1 & \lambda+2 & -\lambda-2 \end{vmatrix} = 0$$

$\therefore \quad (6 - 2\lambda)(2\lambda + 1) = 0$

$\therefore \quad \lambda = 3 \text{ or } -\frac{1}{2}$

But $\lambda = 3$ satisfies all other conditions.

$\therefore \quad \lambda^2 + \lambda = 12$

10. If A, B and $(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))$ are non-singular matrices of same order, then the inverse of $A(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))^{-1} B$, is equal to

(1) $\text{adj}(B^{-1}) + \text{adj}(A^{-1})$

(2) $\frac{1}{|AB|}(\text{adj}(B) + \text{adj}(A))$

(3) $AB^{-1} + A^{-1}B$

(4) $\frac{AB^{-1}}{|A|} + \frac{BA^{-1}}{|B|}$

Answer (2)

Sol. $[A(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))^{-1}B]^{-1}$

$= B^{-1}(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))A^{-1}$

$$= B^{-1}\left[\frac{A}{|A|} + \frac{B}{|B|}\right]A^{-1}$$

$$= \left[\frac{B^{-1}A}{|A|} + \frac{I}{|B|}\right]A^{-1}$$

$$= \frac{B^{-1}}{|A|} + \frac{A^{-1}}{|B|}$$

$$= \frac{\text{adj}(B)}{|AB|} + \frac{\text{adj}(A)}{|AB|}$$

$$= \frac{1}{|AB|}(\text{adj}(B) + \text{adj}(A))$$

11. If the first term of an A.P. is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to

(1) –1080 (2) –1200

(3) –120 (4) –1020

Answer (1)

Sol. Sum of first 4 terms

$= \frac{1}{5} \times$ (sum of next 4 terms)

$$\frac{4}{2}(2a + 3d) = \frac{1}{5}(4a + 22d)$$

$$\Rightarrow \quad 16a = -8d \quad \Rightarrow \quad a = -\frac{d}{2}$$

$a = 3, d = -6$

$$S_{20} = \frac{20}{2}[2(3) + 19(-6)]$$

$= -10(18.6)$

$= -1080$

12. Let the arc AC of a circle subtend a right angle at the centre O. If the point B on the are AC, divides the arc AC such that $\frac{\text{length of arc } AB}{\text{length of arc } BC} = \frac{1}{5}$, and $\overrightarrow{OC} = \alpha\overrightarrow{OA} + \beta\overrightarrow{OB}$, then $\alpha + \sqrt{2}\left(\sqrt{3} - 1\right)\beta$ is equal to

(1) $5\sqrt{3}$ (2) $2 + \sqrt{3}$

(3) $2\sqrt{3}$ (4) $2 - \sqrt{3}$

Answer (4)



Sol. Let the radius be r

B divides the arc in 1 : 5, so AB would subtend an angle of 15° at origin O and similarly, BC would subtend an angle of 75° at origin.

$\overrightarrow{OA}=\vec{a}, \overrightarrow{OB}=\vec{b}, \overrightarrow{OC}=\vec{c}$

$\vec{b}=r(\cos 75°)\hat{i}+r(\sin 75°)\hat{j}$

$$\vec{b}=\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)\vec{c}+\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)\vec{a}$$

$$2\sqrt{2}\vec{b}=\left(\sqrt{3}-1\right)\vec{c}+\left(\sqrt{3}+1\right)\vec{a}$$

$$\overrightarrow{OC}=-\underbrace{\frac{\left(\sqrt{3}+1\right)}{\left|\sqrt{3}-1\right|}}_{\alpha}\overrightarrow{OA}+\underbrace{\frac{2\sqrt{2}}{\left|\sqrt{3}-1\right|}}_{\beta}\overrightarrow{OB}$$

$$\alpha+\sqrt{2}\left(\sqrt{3}-1\right)\beta=2-\sqrt{3}$$

13. If the line $3x - 2y + 12 = 0$ intersects the parabola $4y = 3x^2$ at the points A and B, then at the vertex of the parabola, the line segment AB subtends an angle equal to

(1) $\tan^{-1}\left(\frac{9}{7}\right)$ (2) $\tan^{-1}\left(\frac{11}{9}\right)$

(3) $\frac{\pi}{2}-\tan^{-1}\left(\frac{3}{2}\right)$ (4) $\tan^{-1}\left(\frac{4}{5}\right)$

Answer (1)

Sol. Line $L : 3x - 2y + 12 = 0$

Parabola $P : 4y = 3x^2$

By putting $y=\frac{3x^2}{4}$ in equation of line

We get, $3x-2\left(\frac{3x^2}{4}\right)+12=0$

$\Rightarrow 6x-3x^2+24=0$

$\Rightarrow x^2-2x+8=0$

$\Rightarrow x=4,-2$

for $x = 4$, we get $y = 12$

for $x = –2$, we get $y = 3$

So, points A and B are (4, 12) and (–2, 3)

Now, Vertex of parabola is (0, 0)

$$\Rightarrow \tan\theta=\left|\frac{3-\left(\frac{-3}{2}\right)}{1+3\left(\frac{-3}{2}\right)}\right|$$

$$\tan\theta=\frac{9}{7}$$

$$\Rightarrow \theta=\tan^{-1}\left(\frac{9}{7}\right)$$

$\Rightarrow$ Option (1) is correct

14. Let P be the foot of the perpendicular from the point $Q(10, –3, –1)$ on the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z+1}{-2}$. Then the area of the right-angled triangle PQR, where R is the point (3, –2, 1), is

(1) $3\sqrt{30}$ (2) $9\sqrt{15}$

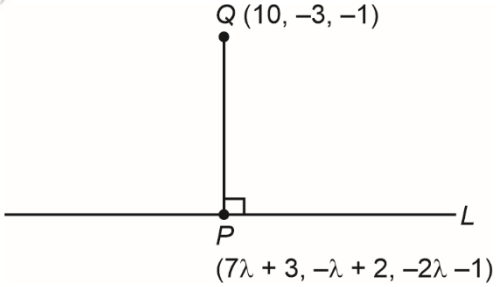
(3) $8\sqrt{15}$ (4) $\sqrt{30}$

Answer (1)

Sol. Q (10, –3, –1)

$$L:\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z+1}{-2}$$

General point on $L(7\lambda+3,-\lambda+2,-2\lambda-1)$



DR's of PQ are $7\lambda - 7, -\lambda + 5, -2\lambda$

$PQ \perp L$

$\Rightarrow (7\lambda-7)7+(-1)(-\lambda+5)+(-2\lambda)(-2)=0$

$\Rightarrow 49\lambda-49+\lambda-5+4\lambda=0$



$\Rightarrow 54\lambda - 54 = 0 \Rightarrow \lambda = 1$

$P(10, 1, -3)$ and $R(3, -2, 1)$

$\overrightarrow{PQ} = -4\hat{j} + 2\hat{k}$, $\overrightarrow{QR} = -7\hat{i} + \hat{j} + 2\hat{k}$,

$\overrightarrow{PR} = -7\hat{i} - 3\hat{j} + 4\hat{k}$

$\overrightarrow{PQ} \cdot \overrightarrow{QR} = 0 \Rightarrow$ right angle at Q

$\Rightarrow \text{ar}(\Delta PQR) = \frac{1}{2} \times PQ \times QR$

$= \frac{1}{2} \times \sqrt{20} \times \sqrt{54}$

$= \frac{1}{2} \times 2 \times 3\left(\sqrt{5} \times \sqrt{6}\right)$

$= 3\sqrt{30}$

$\Rightarrow$ Option (1) is correct

15. The value of $(\sin 70°)(\cot 10° \cot 70° - 1)$ is

(1) 0 (2) 2/3

(3) 3/2 (4) 1

Answer (4)

Sol. $(\sin 70°)(\cot 10° \cot 70° - 1)$

$= \sin 70° \cot 10° \cot 70° - \sin 70°$

$= \cot 10° \cos 70° - \sin 70°$

$= \frac{\cos 10° \cos 70° - \sin 70° \sin 10°}{\sin 10°}$

$= \frac{\cos(10° + 70°)}{\sin 10°}$

$= \frac{\cos 80°}{\sin 10°} = 1$

$\Rightarrow$ Option (4) is correct

16. Let a curve $y = f(x)$ pass through the points $(0, 5)$ and $(\log_e 2, k)$. If the curve satisfies the differential equation $2(3 + y)e^{2x}\,dx - (7 + e^{2x})dy = 0$, then k is equal to

(1) 8 (2) 4

(3) 32 (4) 16

Answer (1)

Sol. $\frac{dy}{dx} = \frac{2(3+y) \cdot e^{2x}}{7 + e^{2x}}$

$\frac{dy}{dx} - \frac{2ye^{2x}}{7 + e^{2x}} = \frac{6 \cdot e^{2x}}{7 + e^{2x}}$

I.F. $= e^{-\int \frac{2e^{2x}}{7+e^{2x}} dx}$

$\Rightarrow e^{-\ln(7+e^{2x})}$

$= \frac{1}{7 + e^{2x}}$

$y \cdot \frac{1}{7 + e^{2x}} = \int \frac{6e^{2x}}{\left(7 + e^{2x}\right)^2} dx$

$\frac{y}{7 + e^{2x}} = \frac{-3}{7 + e^{2x}} + C$

$\therefore y(0) = 5$

$\Rightarrow \frac{5}{8} = \frac{-3}{8} + C$

$\Rightarrow C = 1$

$\therefore y = -3 + 7 + e^{2x}$

$y = e^{2x} + 4$

$\therefore k = 8$

17. If $\frac{\pi}{2} \le x \le \frac{3\pi}{4}$, then $\cos^{-1}\left(\frac{12}{13}\cos x + \frac{5}{13}\sin x\right)$ is equal to

(1) $x - \tan^{-1}\frac{4}{3}$ (2) $x + \tan^{-1}\frac{4}{5}$

(3) $x + \tan^{-1}\frac{5}{12}$ (4) $x - \tan^{-1}\frac{5}{12}$

Answer (4)

Sol. $\frac{12}{13}\cos x + \frac{5}{13}\sin x$

Let $\tan\alpha = \frac{5}{12}, \alpha \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow \sin\alpha = \frac{5}{13}, \cos\alpha = \frac{12}{13}$



$\Rightarrow \frac{12}{13}\cos x + \frac{5}{13}\sin x = \cos\alpha \cos x + \sin\alpha \sin x$

$= \cos(x - \alpha)$

$\Rightarrow \cos^{-1}[\cos(x - \alpha)] = x - \alpha$

$= x - \tan^{-1}\left(\frac{5}{12}\right)$

18. Marks obtains by all the students of class 12 are presented in a frequency distribution with classes of equal width. Let the median of this grouped data be 14 with median class interval 12-18 and median class frequency 12. If the number of students whose marks are less than 12 is 18, then the total number of students is

(1) 40 (2) 52

(3) 48 (4) 44

Answer (4)

Sol. $M = L + \frac{\frac{n}{2} - Cf}{f} \times h$

$14 = 12 + \frac{\frac{n}{2} - 18}{12} \times 6$

$2 \times 2 = \frac{n}{2} - 18$

$\frac{n}{2} = 4 + 18$

$n = 44$

19. If the function

$$f(x) = \begin{cases} \frac{2}{x}\{\sin(k_1+1)x + \sin(k_2-1)x\}, & x < 0 \\ 4, & x = 0 \\ \frac{2}{x}\log_e\left(\frac{2+k_1x}{2+k_2x}\right), & x > 0 \end{cases}$$

is continuous at $x = 0$ then $k_1^2 + k_2^2$ is equal to

(1) 20 (2) 5

(3) 8 (4) 10

Answer (4)

Sol. $\lim\limits_{x\to0^-} f(x) = \lim\limits_{x\to0^-} 2\left(\frac{\sin(k_1+1)x}{x} + \frac{\sin(k_2-1)x}{x}\right)$

$= 2((k_1 + 1) + (k_2 - 1))$

$= 2(k_1 + k_2)$ ...(i)

$\lim\limits_{x\to0^+} f(x) = \lim\limits_{x\to0^+} 2\left(\frac{\ln\left(1+\frac{k_1}{2}x\right)}{x} - \frac{\ln\left(1+\frac{k_2}{2}x\right)}{x}\right)$

$= k_1 - k_2$ ...(ii)

For continuity at $x = 0$

$2(k_1 + k_2) = 4$ and $k_1 - k_2 = 4$

$\Rightarrow k_1 + k_2 = 2$ and $k_1 - k_2 = 4$

$\Rightarrow k_1 = 3$ & $k_2 = -1$

$k_1^2 + k_2^2 = 10$

20. The value of

$$\int_{e^2}^{e^4} \frac{1}{x}\left(\frac{e^{\left((\log_e x)^2+1\right)^{-1}}}{e^{\left((\log_e x)^2+1\right)^{-1}} + e^{\left((6-\log_e x)^2+1\right)^{-1}}}\right)dx \text{ is}$$

(1) 1 (2) e^2

(3) 2 (4) $\log_e 2$

Answer (1)

Sol. Put $\ln x = t \Rightarrow \frac{1}{x}dx = dt$

x	t
e^2	2
e^4	4

$$I = \int_2^4 \frac{e^{(t^2+1)^{-1}}}{e^{(t^2+1)^{-1}} + e^{\left((6-t)^2+1\right)^{-1}}}dt \quad ...(i)$$

$$I = \int_2^4 \frac{e^{\left((6-t)^2+1\right)^{-1}}}{e^{\left((6-t)^2+1\right)^{-1}} + e^{(t^2+1)^{-1}}}dt \quad ...(ii)$$

$\left\{\text{Using}\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx\right\}$

Adding (i) and (ii) gives

$2I = \int dt \Rightarrow I = 1$



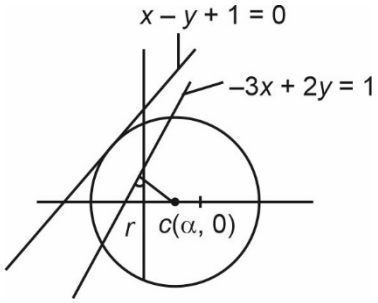
SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. Let the circle C touch the line $x - y + 1 = 0$, have the centre on the positive x-axis, and cut off a chord of length $\frac{4}{\sqrt{13}}$ along the line $-3x + 2y = 1$. Let H be the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$, whose one of the foci is the centre of C and the length of the transverse axis is the diameter of C. Then $2\alpha^2 + 3\beta^2$ is equal to _____.

Answer (19)

Sol.



$$r = \left|\frac{a+1}{\sqrt{2}}\right| \Rightarrow (a+1)^2 = 2r^2$$

Also $\left(\frac{3a-1}{\sqrt{13}}\right)^2 + \left(\frac{2}{\sqrt{13}}\right)^2 = r^2$

$$\Rightarrow \left(\frac{3a-1}{\sqrt{13}}\right)^2 + \frac{4}{13} = \frac{(a+1)^2}{2}$$

$5a^2 - 14a - 3 = 0$

$\therefore \quad a = -\frac{1}{5}, 3$

$\because \ a \neq -\frac{1}{5} \Rightarrow a = 3$

$\Rightarrow \quad r = 2\sqrt{2}$

$\because$ One focus of $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ is $(3, 0)$

$\Rightarrow \quad \alpha e = 3$ and $2\alpha = 4\sqrt{2}$

$\Rightarrow \quad \alpha = 2\sqrt{2} \Rightarrow \alpha^2 = 8$

$$\alpha^2\left[1 + \frac{\beta^2}{\alpha^2}\right] = 9$$

$\alpha^2 + \beta^2 = 9$

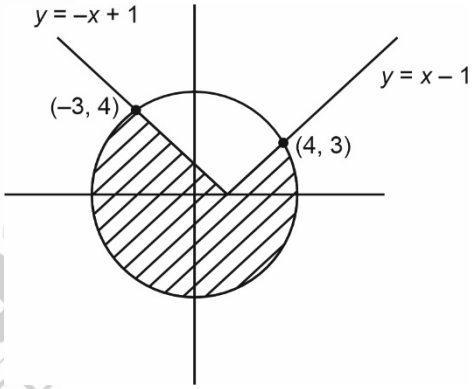
$\Rightarrow \quad \beta^2 = 1$

$\therefore \quad 2\alpha^2 + 3\beta^2 = 19$

22. If the area of the larger portion bounded between the curves $x^2 + y^2 = 25$ and $y = |x - 1|$ is $\frac{1}{4}(b\pi + c)$, $b, c \in \mathbb{N}$, then $b + c$ is equal to _____.

Answer (77)

Sol.



Area of shaded region

$$= 25\pi - \left[\int_{-3}^{4}\left(\sqrt{25 - x^2} - |x - 1|\right)dx\right]$$

$$= 25\pi - \left[\frac{1}{2}\sqrt{25 - x^2} + \frac{25}{2}\sin^{-1}\frac{x}{5}\right]_{-3}^{4} + \left(8 + \frac{9}{2}\right)$$

$$= \frac{1}{4}[75\pi + 2]$$

$\Rightarrow \quad b = 75, c = 2$

$\therefore \quad b + c = 77$



23. The sum of all rational terms in the expansion of $\left(1+2^{\frac{1}{3}}+3^{\frac{1}{2}}\right)^6$ is equal to _____

Answer (612)

Sol. The general term of multinomial expansion is

$$\frac{6!}{\alpha!\,\beta!\,\gamma!}(1)^{\alpha}\left(2^{\frac{1}{3}}\right)^{\beta}\left(3^{\frac{1}{2}}\right)^{\gamma}$$

For terms to be rational $3|\beta$ and $2|\gamma$

β	γ	α	Term
0	0	6	1
0	2	4	$15.3 = 45$
0	4	2	$15.3^2 = 135$
0	6	0	$1.3^3 = 27$
3	0	3	$20.2 = 40$
3	2	1	$60.2.3 = 360$
6	0	0	$1.4 = 4$

$\Rightarrow$ Sum of rational terms

$= 1 + 45 + 135 + 27 + 40 + 360 + 4 = 612$

24. If the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ has equal roots, where $a + c = 15$ and $b = \frac{36}{5}$, then $a^2 + c^2$ is equal to _____.

Answer (117)

Sol. Clearly one root is 1, another root is also 1.

Product of roots = 1

$$\frac{c(a-b)}{a(b-c)} = 1$$

$c(a-b) = a(b-c)$

$ac - bc = ab - ac$

$2ac = b(a+c)$

$$2ac = \frac{36}{5}(15)$$

$ac = 18 \times 3 = 54$

$\therefore\ a^2 + c^2 = (a+c)^2 - 2ac$

$= (15)^2 - 2(54)$

$= 225 - 108$

$= 117$

25. If the set of all values of a, for which the equation $5x^3 - 15x - a = 0$ has three distinct real roots, is the interval (α, β), then $\beta - 2\alpha$ is equal to _____.

Answer (30)

Sol. $5x^3 - 15x - a = 0$ has 3 distinct real solution

$\therefore$ Let $f(x) = 5x^3 - 15x - a$

$f'(x) = 15x^2 - 15 = 0$

$15x^2 = 15$

$x^2 = 1$

$\boxed{x = \pm 1}$

$\therefore\ f(1)\, f(-1) < 0$

$(a - 10)(a + 10) < 0$

$a \in (-10, 10)$

$\therefore\ \beta - 2\alpha = 10 + 2(10) = 30$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. A solid sphere of mass 'm' and radius 'r' is allowed to roll without slipping from the highest point of an inclined plane of length 'L' and makes an angle 30° with the horizontal. The speed of the particle at the bottom of the plane is v_1. If the angle of inclination is increased to 45° while keeping L constant. Then the new speed of the sphere at the bottom of the plane is v_2. The ratio $v_1^2 : v_2^2$ is

(1) 1 : 3

(2) $1:\sqrt{2}$

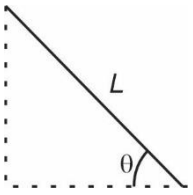
(3) 1 : 2

(4) $1:\sqrt{3}$

Answer (2)

Sol. Loss in P.E. = gain in K.E.

$$mgL\sin\theta = \frac{1}{2}\left(\frac{7}{5}\right)mv^2$$



$$\Rightarrow \quad v^2 \propto \sin\theta$$

$$\frac{v_1^2}{v_2^2} = \frac{\sin 30^\circ}{\sin 45^\circ} = \frac{1}{\sqrt{2}}$$

27. A gun fires a lead bullet of temperature 300 K into a wooden block. The bullet having melting temperature of 600 K penetrates into the block and melts down. If the total heat required for the process is 625 J, then the mass of the bullet is _____ grams.

(Latent heat of fusion of lead = 2.5×10^4 JKg^{-1} and specific heat capacity of lead 125 JKg^{-1} K^{-1})

(1) 20 (2) 5

(3) 10 (4) 15

Answer (3)

Sol. $Q = ms\Delta T + mL$

$625 = m \times 125 \times 300 + m \times 2.5 \times 10^4$

$625 = m\{3.75 + 2.5\} \times 10^4$

$$\Rightarrow \frac{625}{6.25} \times 10^{-4} \text{ kg} = 10\text{ g} = m$$

28. A radioactive nucleus n_2 has 3 times the decay constant as compared to the decay constant of another radioactive nucleus n_1. If initial number of both nuclei are the same, what is the ratio of number of nuclei of n_2 to the number of nuclei of n_1, after one half-life of n_1?

(1) 1/8 (2) 1/4

(3) 4 (4) 8

Answer (2)

Sol. $\lambda_2 = 3\lambda \qquad \lambda_1 = \lambda$

$N = N_0 e^{-\lambda t}$

$N_1 = N_0 e^{-\lambda t}$

$N_2 = N_0 e^{-3\lambda t}$

$$\frac{N_2}{N_1} = e^{-2\lambda t}$$

